CS 1470 · DEEP LEARNING · FINAL PROJECT

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Multi-Agent Reinforcement Learning in Pommerman: *Comparing PPO and Dreamer-Style World Models*

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SECTION 01

Introduction

Reinforcement learning has achieved strong results in many single-agent environments, where an agent learns from relatively stable dynamics and a fixed objective. Real-world decision-making, however, often involves multiple agents whose actions change the environment and each other's learning problem.

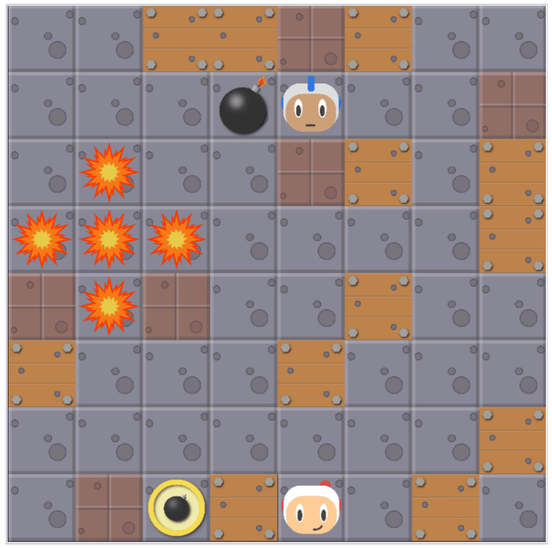
This project studies multi-agent reinforcement learning in Pommerman, a competitive grid-world environment requiring planning, adaptation, and interaction. We compare PPO with Dreamer-style world model agents to better understand how model-free and model-based RL methods behave in a more complex multi-agent setting.

SECTION 02

Game Environment

Pommerman is a multi-agent grid-world environment inspired by Bomberman, introduced by Resnick et al. in *Pommerman: A Multi-Agent Playground*. Agents move through a maze, place bombs, collect power-ups, and try to survive while eliminating opponents.

It is a useful testbed for multi-agent reinforcement learning because success requires both short-term reactions and longer-term planning: agents must avoid explosions, navigate around walls, anticipate opponent behavior, and decide when to act aggressively or defensively. Its combination of discrete actions, spatial observations, sparse rewards, and direct agent interaction makes it more challenging than many standard single-agent RL benchmarks while still being compact enough for controlled experimentation.

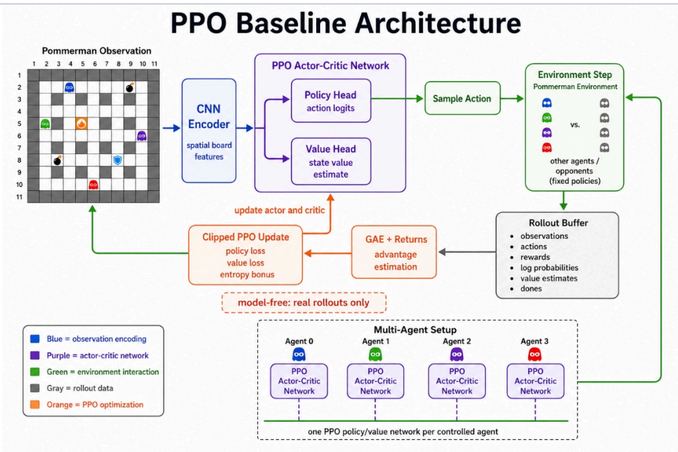


[FIGURE 1] The Pommerman environment: agents navigate a grid of destructible wooden walls and indestructible rigid walls, placing bombs that explode after a short fuse to clear walls and eliminate opponents.

SECTION 03

Architecture

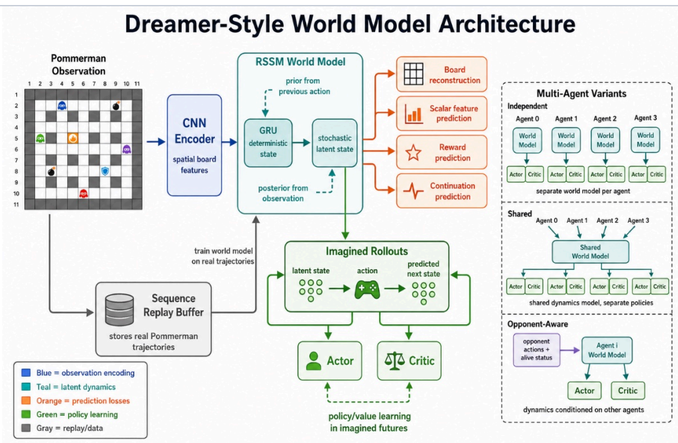
Our model suite compares a model-free PPO baseline against several Dreamer-style world model agents. The PPO agents use a convolutional encoder over Pommerman board observations, followed by separate policy and value heads trained with clipped policy updates and generalized advantage estimation. The Dreamer-style agents instead learn a recurrent state-space world model: observations are encoded with a CNN, passed through a GRU-based latent dynamics model, and decoded to predict future board states, rewards, and episode continuation. Policies and critics are then trained on imagined rollouts from this learned latent model.



PPO Baseline

MODEL-FREE

CNN encoder → actor-critic heads → clipped PPO update with GAE. One policy/value network per controlled agent; real rollouts only.



Dreamer-Style World Model

MODEL-BASED

CNN → RSSM (GRU + stochastic latent) predicts board, reward, and continuation. Actor/critic train on imagined rollouts.

Multi-Agent Variants

To study multi-agent structure, we implemented three world model strategies:

VARIANT A

Independent

Each player has its own world model and policy.

VARIANT B

Shared

Agents share a common world model but keep separate actor-critic heads.

VARIANT C

Opponent-Aware

Each agent's dynamics model is conditioned on other agents' recent actions and alive status.

SECTION 05

Challenges

One of the biggest challenges was making Dreamer-style training stable in a multi-agent setting. Unlike PPO, the Dreamer variants depend on learning a usable model of the environment before policy learning can really improve, which is difficult in Pommerman because the model has to predict bomb dynamics, board changes, rewards, and opponent behavior.

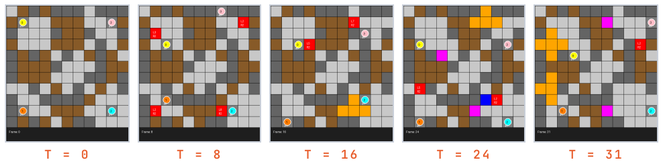
To address this, we added sequence-aware replay for recurrent world-model training, gated updates on valid same-episode trajectories, and tested independent, shared, and opponent-aware world model variants. We also moved beyond sparse win/loss rewards by adding shaped rewards that give agents credit for useful events like destroying wood, collecting power-ups, eliminating opponents, and placing useful bombs, while penalizing inactive or unproductive behavior such as unnecessary stopping, blocked moves, wasted bombs, ties, and long episodes. This helped push training away from passive survival and toward more purposeful interaction with the environment.

SECTION 06

Results

Our short-budget Pommerman experiments showed a clear gap between the model-free PPO baseline and the Dreamer-style world-model agents. PPO was the most stable approach, reaching an evaluation mean reward of **1.14** and a **50% win rate** in the main 2048-step study. The initial Dreamer variants, including independent, shared, and opponent-aware models, learned parts of the environment dynamics but did not translate those predictions into reliable control, with best win rates still at **0%** in the original three-way comparison. After additional debugging and training improvements, the shared Dreamer variant achieved nonzero performance, with the best optimized checkpoint reaching about **17.2% win rate** over 64 evaluation episodes. Overall, our results suggest that PPO remains the stronger short-horizon baseline, while Dreamer-style world modeling is promising but substantially harder to train reliably in multi-agent Pommerman.

METHOD	WIN RATE
PPO (baseline)	50%
Dreamer · Independent	~0%
Dreamer · Shared	17.2%
Dreamer · Opponent-Aware	~0%



[FIGURE 2] Dreamed latent rollout of the shared Dreamer agent (yellow, ID 0) on Pommerman. Frames sampled across an evaluation episode show the world model tracking board state, opponents, and the agent's own movement decisions over time.

SECTION 07

Discussion & Future Work

These results highlight the difficulty of extending world model-based reinforcement learning to multi-agent settings. While Dreamer-style methods are appealing because they can learn from imagined futures, our experiments suggest that learning an accurate model of Pommerman is not enough on its own: the agent must also turn that model into effective strategic behavior.

The stronger PPO performance shows that model-free methods can be more stable under limited compute, while the Dreamer results reveal useful directions for future work, including **longer training runs**, **more seeds**, **improved reward shaping**, **stronger opponent modeling**, and **diagnostics for whether imagined rollouts match real game dynamics**.

REFERENCES

[1] Resnick, C., Eldridge, W., Ha, D., Britz, D., Foerster, J. N., Togelius, J., Cho, K., & Bruna, J. (2018). *Pommerman: A Multi-Agent Playground*. AIIDE Workshops. arXiv:1809.07124. arxiv.org/abs/1809.07124

[2] Hafner, D., Lillicrap, T. P., Ba, J., & Norouzi, M. (2020). *Dream to Control: Learning Behaviors by Latent Imagination*. ICLR. arXiv:1912.01603. arxiv.org/abs/1912.01603